



Week 5 Lecture 5

▼ Class	BSCCS2003
🕒 Created	@October 4, 2021 1:13 AM
🔗 Materials	
# Module #	31
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Routes and Controllers

Web applications

- Client-Server model
- Stateless: Server does not know the present state of the client
 - Must be ready to respond to whatever the client requests without assuming anything about the client
- Requests sent through HTTP protocol
 - Use variants of the GET, POST (verbs) to convey meaning
 - Use URL (Uniform Resource Locator) structure to convey context

Routing: mapping URLs to actions

Python Decorators

- Add extra functionality on top of a function
- "@" - decorators before function name
- Effectively, it is a function of a function
 - Take the new inner function as an argument
 - Return a function that does something before calling the inner function

Basic routing in Flask

```
from flask import Flask
app = Flask(__name__)
```

```
@app.route('/')
def home():
    return 'Hello, World!'
```

HTTP verbs

```
@app.route('/', methods=['GET'])
def index():
    ...

@app.route('/create', methods=['POST'])
def store():
    ...
```

CRUD-like functionality

```
# Assume 'index', 'store' etc. are functions

@app.route('/', method=['GET'])(index)
@app.route('/create', method=['POST'])(store)
@app.route('/<int:user_id>', method=['GET'])(show)
@app.route('/<int:user_id>/edit', method=['POST'])(update)
@app.route('/<int:user_id>', method=['DELETE'])(destroy)
```

Summary

- Flask is not natively MVC
 - But MVC is more a way of thought than a framework
- Simple URL routing
- Helpers to do large scale routing of common functions

Structure the application with separation of concerns in mind

- MVC is just one way to achieve clean design